

Walkability and Typical Home Values Across U.S. ZIP Codes

Research question: In places where daily errands are easier to do on foot, are typical home values higher?
Data: EPA National Walkability Index (ZIP) merged with Zillow-style typical home values (``zillow_clean.csv``).

- Cleaning highlights:
- ZIP codes standardized to 5 digits for a reliable merge.
 - Rows with missing walk score, price, or state removed (0 rows).
 - Extreme home values trimmed at the 1st–99th percentile to limit outlier leverage while keeping most of the market (68,073–1,425,633).
 - Walk scores trimmed at the 1st–99th percentile for robustness.

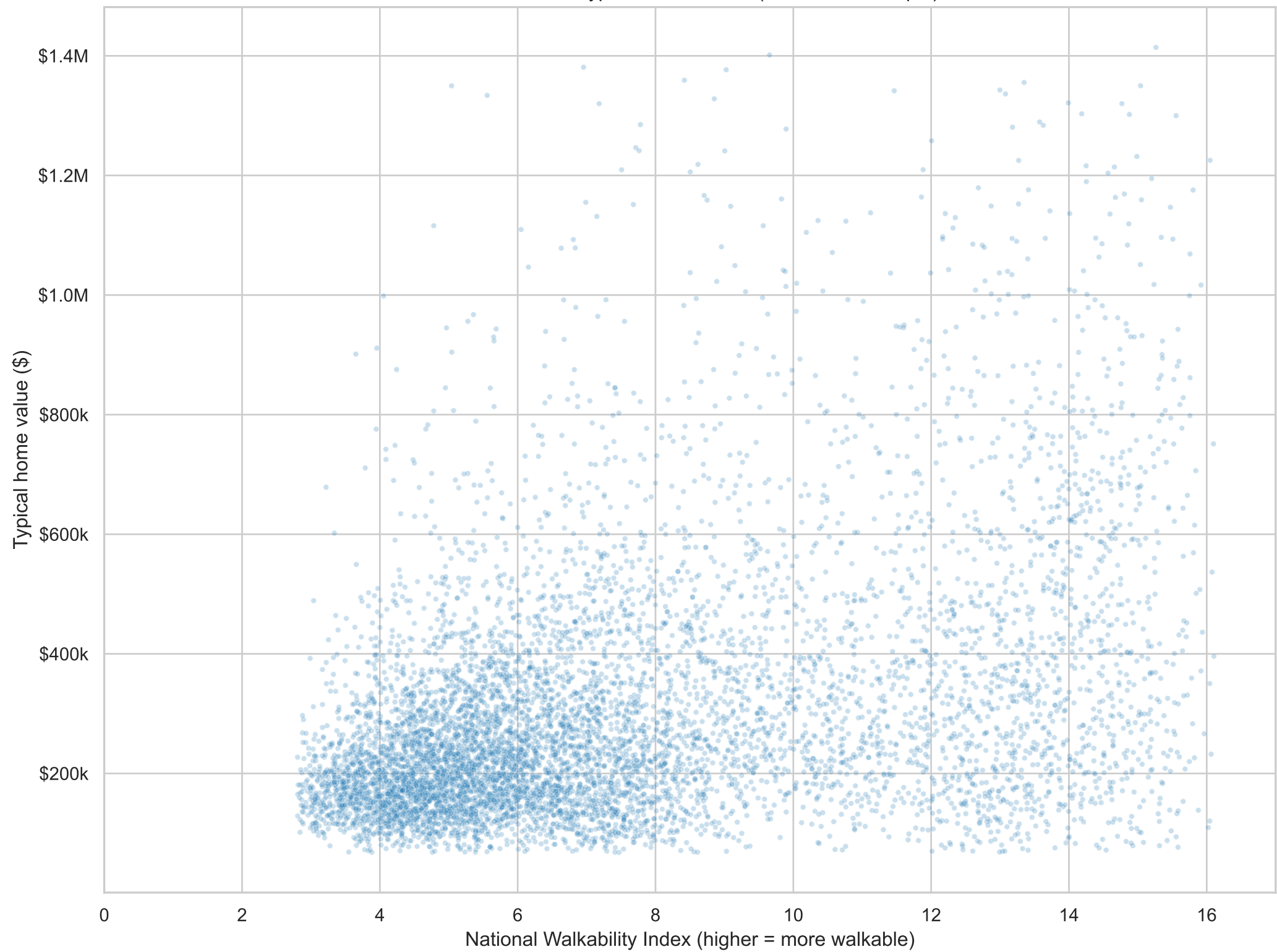
Final analytic sample: 18,535 ZIP codes after cleaning.

Why it matters: Walkability is a proxy for land-use intensity, amenities, and transport options. If prices rise with walkability, that signals how households trade off housing costs against neighborhood accessibility—useful for planners, buyers, and researchers studying urban form and affordability.

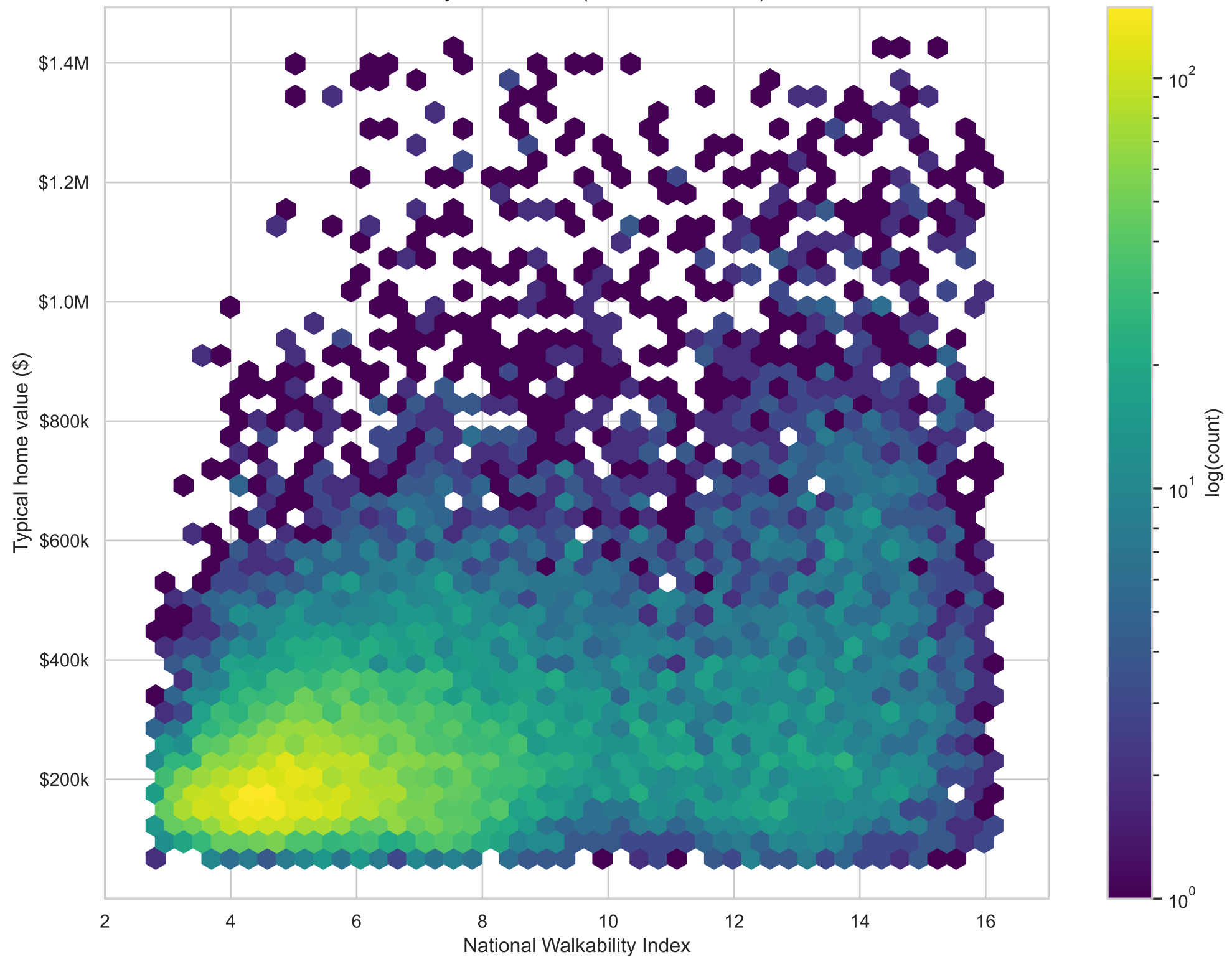
Summary statistics (ZIP-level, cleaned sample)

	NatWalkInd	home_value
count	18535.0	18535.0
mean	7.58	303321.87
std	3.38	203695.82
min	2.81	68073.46
5%	3.61	108063.44
25%	4.93	168508.18
50%	6.55	240619.27
75%	9.64	367222.56
95%	14.26	720676.56
max	16.12	1425633.21

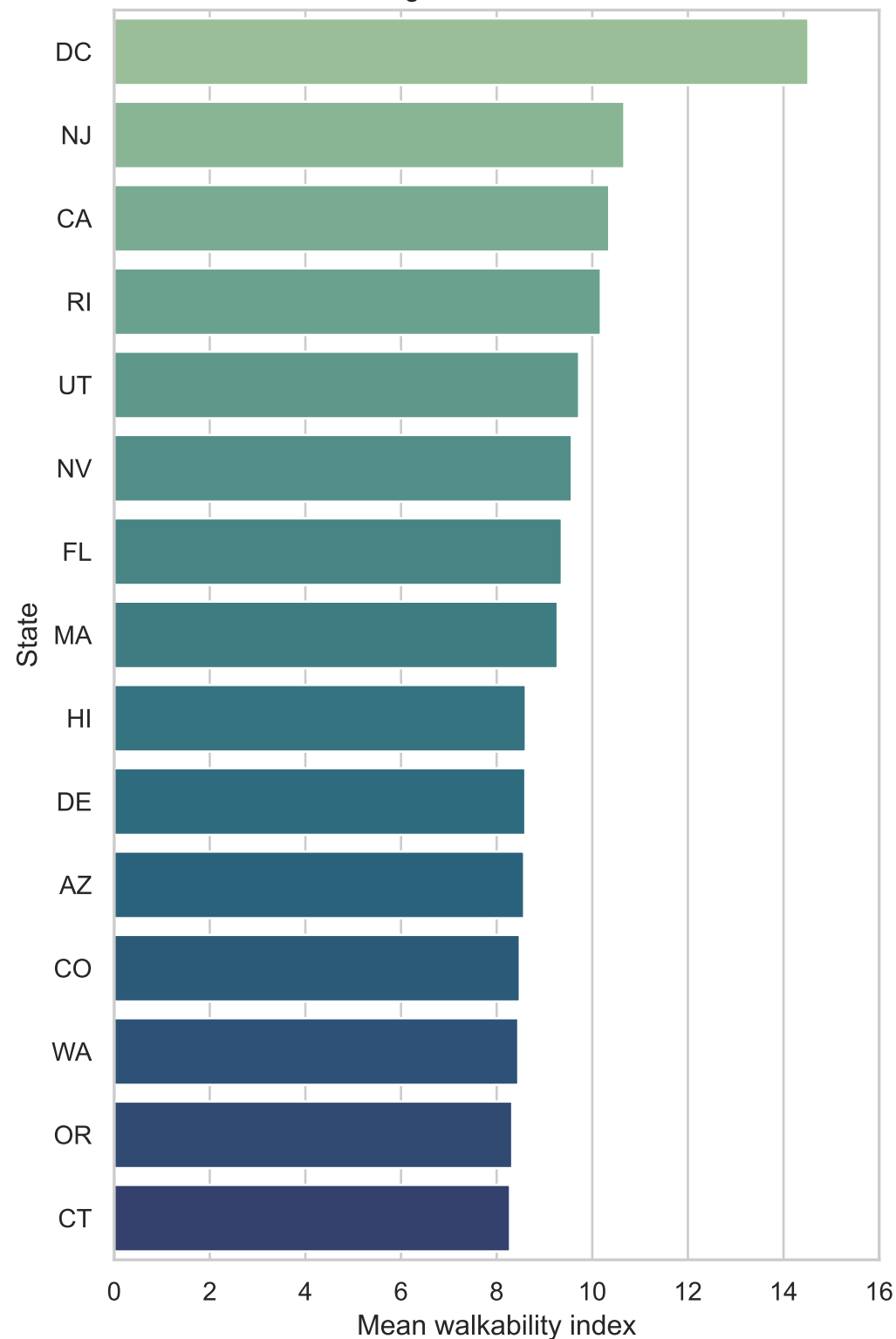
Walk Score vs. Typical Home Value (random subsample)



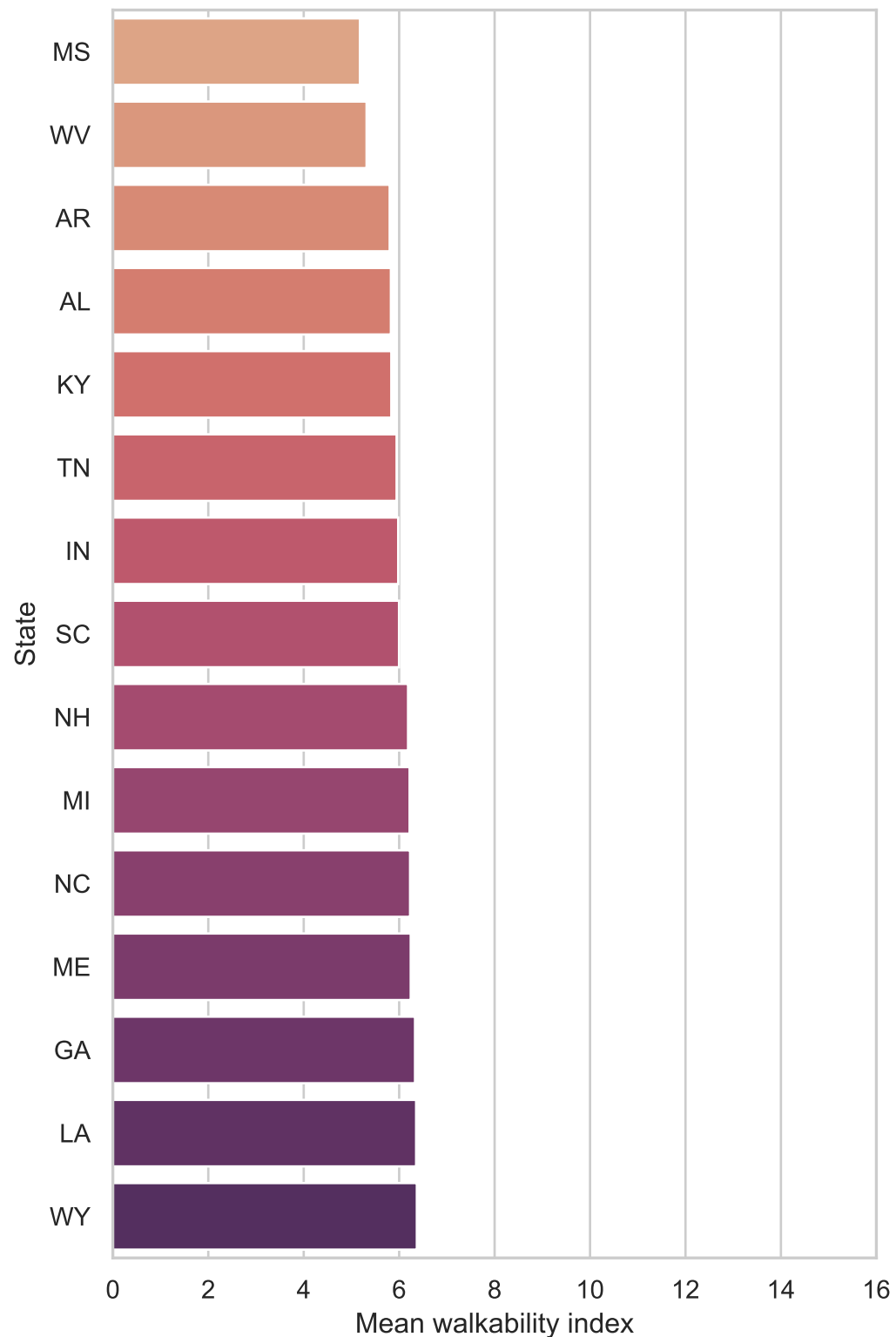
Density of ZIP Codes (darker = more ZIPs)



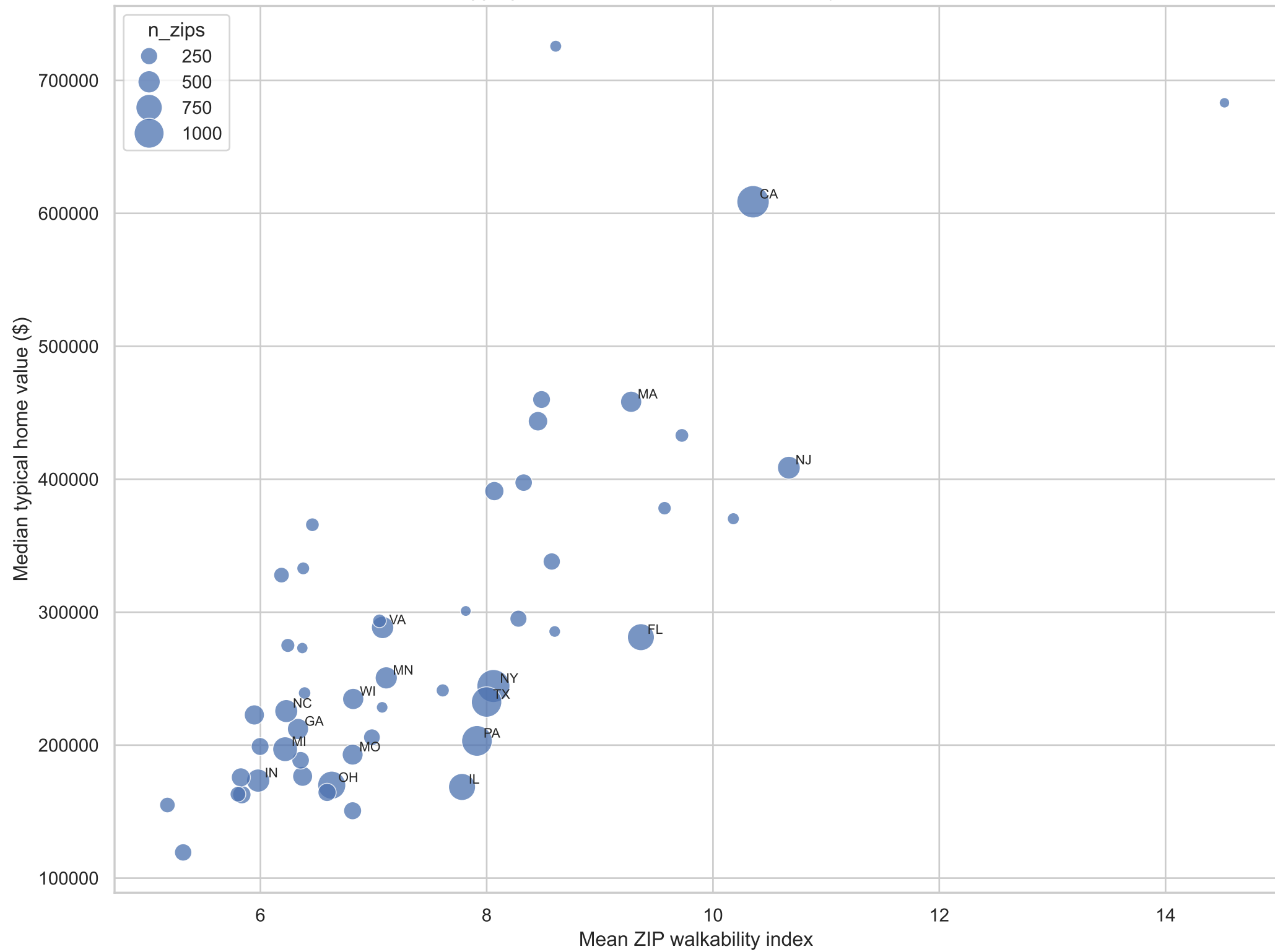
States with highest mean ZIP walk scores



States with lowest mean ZIP walk scores



State aggregates: mean walk score vs. median typical home value



Residuals from simple OLS: log home value ~ walk score



Statistical analysis (ZIP-level, cleaned sample)

Pearson correlation (walk, price): $r = 0.394$, $p = 0.00e+00$

Spearman rank correlation: $\rho = 0.372$, $p = 0.00e+00$

Home values are right-skewed; log transform stabilizes variance.

Pearson correlation (walk, $\log_{10}(\text{price})$): $r = 0.378$, $p = 0.00e+00$

Simple OLS: $\log_{10}(\text{home_value}) = \beta_0 + \beta_1 * \text{NatWalkInd}$

β_0 (intercept): 11.9653

β_1 (per walk unit): 0.06377

R-squared on log scale: 0.143

Interpretation: A positive association means ZIPs with higher walk scores tend to have higher typical prices, but correlation is not causation—dense, high-amenity places can be both walkable and expensive for many reasons.